Transportation Research Part C 127 (2021) 103132

Contents lists available at ScienceDirect

Transportation Research Part C

journal homepage: www.elsevier.com/locate/trc

The potential impact of media commentary and social influence on consumer preferences for driverless cars

Milad Ghasri[a,*], Akshay Vij[b]

[a] School of Engineering and IT, UNSW Canberra, ACT 2612, Australia
[b] Institute for Choice, University of South Australia, Adelaide, SA 5000, Australia

ARTICLE INFO

Keywords:
Autonomous vehicles (AV)
Driverless cars
Word of mouth
Media sentiments
Market penetration
Social influence

ABSTRACT

The emerging technology of autonomous vehicles (AVs) is envisaged to substantially alter transport systems and travel behaviour. However, this impact is conditional on the level and rate of AV market penetration. This study examines consumers' consideration to adopt AVs, with a particular focus on the impact of informational cues from different communication channels. Data for our analysis is collected through an online survey conducted in December 2019, and our sample comprises 862 residents from Sydney, Australia. The survey asks respondents about their social networks, and their use of different mass and social media platforms. Respondents are subsequently presented a discrete choice experiment where we systematically vary informational cues about AVs from each of these communication channels, and respondents are asked to indicate their willingness to consider AVs under each scenario. This data is used to develop a latent class choice model of AV consideration as a function of the informational cues received from different communication channels. Our model also considers the level of trust that respondents associate with their social contacts when modelling the effect of word-of-mouth from these social contacts. The results reveal eleven segments in the sample that differ from one another in terms of sociodemographic attributes, evaluation of social contact trust, and the relative importance they attach to different communication channels. The findings show social media sentiment has the highest effect on AV consideration for 90 per cent of the sample. Also 3.8 per cent of the sample never consider adopting AVs, regardless of the informational cues.

1. Introduction

The third millennium has marked a significant leap in technological advancement, with substantial impacts on human life. In the field of transport, innovations such as online navigation, adaptive cruise control, ridesharing and car sharing modes, and on-demand services have changed travel behaviour patterns in the last few decades (e.g. Piccinini et al., 2014, Vij et al., 2020, Henao, 2017). Emerging autonomous vehicles (AVs) are envisaged to substantially alter transport systems and travel behaviour even further (e.g. Fagnant and Kockelman, 2015). However, their short and long-term impacts will depend on their rate of adoption and diffusion.

The decision to adopt an innovation is influenced by, among other factors, the information received from various communication channels about the innovation (Rogers, 2010). Information received from communication channels can raise awareness about an innovation at the knowledge formation stage (Garcia, 2005, Rogers, 2010), it can influence the formation of positive or negative

* Corresponding author.
E-mail address: m.ghasri@unsw.edu.au (M. Ghasri).

https://doi.org/10.1016/j.trc.2021.103132
Received 3 January 2021; Received in revised form 28 March 2021; Accepted 6 April 2021
Available online 29 April 2021

attitudes towards the innovation (Weenig and Midden, 1991), and it can influence the decision to adopt or reject the innovation (Shabanpour et al., 2018b, Ghasri et al., 2019).

Communication channels can comprise both mass media and inter-personal networks (Bass, 1969, Rogers, 2010). Mass media channels may include radio, television, newspapers, etc., where information flows from one source to many audiences (Bass, 1969). Inter-personal channels refer to the exchange of information between two or more individuals (Granovetter, 1973). Mass media communication channels are typically more important during the early stages of innovation diffusion. They are essential in creating awareness about the innovation, and have higher impacts on early adopters (Bass, 1969, Rogers, 2010). In contrast, later adopters are usually more reliant on inter-personal communication channels. They tend to depend on verbal and non-verbal confirmation from social contacts before deciding to adopt an innovation (Rogers, 2010). Recent advancements in digital communication have created new channels of communication, such as social media, that have blurred the line between mass media and inter-personal networks, and have emerged as a major source of further influence on the diffusion process (e.g. Erdoğmuş and Cicek, 2012, Lipsman et al., 2012).

Multiple studies have sought to understand public attitudes and perceptions towards AVs (Cunningham et al., 2018, Hulse et al., 2018, Menon et al., 2016, Kyriakidis et al., 2015, Payre et al., 2014, Rödel et al., 2014, Schoettle and Sivak, 2014a, Casley et al., 2013, Sundareswara et al., 2013, Liu, 2020). Commonly identified perceived benefits include greater safety, better fuel economy, more productive use of travel time, and improved accessibility. Conversely, commonly identified perceived barriers include risk of equipment failure, fear of relinquishing control, data privacy issues, and threat from online hackers. However, none of these studies has examined the impact of information received from different communication channels on the formation of consumer preferences for AVs.

To that end, this study models individual willingness to adopt AVs as a function of the information received from different communication channels. We design a survey that asks respondents about their interpersonal networks and their use of different mass and social media platforms. The survey comprises a discrete choice experiment (DCE) where we systematically manipulate informational signals about AVs from different channels across multiple hypothetical scenarios, and we elicit respondents' willingness to adopt AVs under each scenario. The survey is administered online to a sample of 862 residents in Sydney, Australia. The data thus collected is used to develop a latent class choice model (LCCM) framework that is able to measure the impact of different informational signals from different communication channels on respondents' willingness to adopt AVs, and compare how these impacts vary across different respondents as a function of their demographic characteristics. Special focus is given to understanding the differential impact of positive and negative word-of-mouth (WOM) feedback from different social contacts, while accounting for the strength of the relationship between the respondent and each social contact using the concept of trust.

The rest of this paper is organised as follows. Section 2 provides a brief review of the literature of social influence, and AV market uptake studies in the field of transport. Section 3 presents the survey instrument and Section 4 presents the details of the collected sample. Section 5 puts forward an econometric framework to analyse the collected data, and Section 6 presents the parameter estimates. The findings are discussed in detail in Section 7, and the paper is concluded by summarising the highlights of the study in Section 8.

2. Literature review

Social interaction and its impact on individual behaviour is studied from various aspects including social cooperation, social influence, social learning, and social capital (Maness, 2020). Social cooperation examines the active coordination of activities with respect to other members of the household or society (Carrasco and Miller, 2009, Habib and Carrasco, 2011). Social learning and social influence investigates how behaviour is affected by others' opinion or action (Dugundji and Walker, 2005, Axsen et al., 2013), and social capital explains how social networks can empower individuals with their achievements (Maness, 2020, Maness, 2017a). This paper contributes to the social influence aspect by drawing upon the theory of diffusion of innovation (DOI) in the context of autonomous vehicle adoption.

Diffusion of innovation is defined as the spread of innovations through communication channels in a social system over time (Rogers, 2010). The role of communication channels is to transmit information about the innovation within the social system (Mahajan, 2010). The effect of communication channels can vary depending on the channel, the adopter characteristics, and the stage of innovation decision process. Information channels are often divided into mass media and interpersonal communication channels (Mahajan, 2010, Bass, 1969). Mass media channels can reach a large audience rapidly. This channel is often important in creating knowledge about an innovation, especially at the early stages of the innovation diffusion. Interpersonal channels involve a direct exchange of information between two or more individuals. Recent advances in information and communication technologies have created new channels of communication, such as social media, that have blurred the line between mass media and inter-personal networks. Communication channels are essential in forming or changing strongly held attitudes towards innovations (Rogers, 2010).

In the field of transport, social influence and its impact on travel decisions are studied in various contexts including technology adoption (Axsen et al., 2013), activity choice (Maness, 2017b, Kim et al., 2018, Axhausen, 2005), and travel mode choice (Bartle et al., 2013, Sherwin et al., 2014, Páez and Scott, 2007). While the focus in some studies is limited to the social influence among a specific cohort of social contacts (Bartle et al., 2013), in some other studies the boundaries of social network is extended to include friends, peers and colleagues (Sherwin et al., 2014) and those with whom the decision-maker has a weak tie (Maness, 2017b). Regarding the communication channels and their impact on individuals' decision, some researchers have directly studied interpersonal communication and its role in disseminating factual and normative information (Bartle et al., 2013, Axsen et al., 2013), and some have studied the impact of non-verbal observations from the behaviour of the society in terms of market penetration of an innovation (Ghasri et al., 2019, Shabanpour et al., 2018b).

The common finding of all these studies is that individuals' decision is not just dependent on the intrinsic values of the choice, but it is also nudged by social influence and other market externalities. Depending on the mechanism of nudge, theories that explain social influence on consumer behaviour can be divided into two groups. In the first category, social influence is considered as a source of information that can positively or negatively affect the likelihood of adoption. For instance, in DOI, social contacts (along with mass media) can provide knowledge about an innovation, and peers adoption rate is assumed to send a confirmation signal to decision-makers (Rogers, 2010). The signal from previous adopters is not always positive and if previous users are not satisfied with the innovation their feedback may reduce the likelihood of adoption. This concept is addressed as Positive WOM (PWOM) and Negative WOM (NWOM) (East et al., 2016). The second group of theories pertains to signals that the *decision-maker* will communicate back to the social network by adopting or rejecting an innovation. This phenomenon is discussed under different concepts including symbolic consumption (Kim and Jang, 2017), conformity (Asch, 1956, Axsen and Kurani, 2012) and vanity (Grilo et al., 2001). Theories in this category are more applicable to consuming conspicuous goods.

There is an extensive literature on studies assessing consumers' reaction to AVs (Gkartzonikas and Gkritza, 2019, Becker and Axhausen, 2017, Rashidi et al., 2020). Previous researchers studied the impact of awareness (Schoettle and Sivak, 2014b), intention (Panagiotopoulos and Dimitrakopoulos, 2018), attitudes (Asgari and Jin, 2019), beliefs (König and Neumayr, 2017) and confidence in beliefs (Sanbonmatsu et al., 2018). These studies show perceived usefulness, perceived ease to use, and perceived trust are positively correlated with the intention to use AVs (Panagiotopoulos and Dimitrakopoulos, 2018), drivers with negative views towards AVs are least educated about this technology (König and Neumayr, 2017), and there is a psychological barrier towards adopting AVs in some demographics (Sanbonmatsu et al., 2018). Other reported barriers include vehicle cost, liability, software failure and security, safety and reluctance to relinquish control of car (Fagnant and Kockelman, 2015, Kong and Hardman, 2019).

Surveys using stated choice experiments are mainly designed to elicit consumer preference and their willingness to pay for various features of AVs (Shin et al., 2015, Shabanpour et al., 2018a, Daziano et al., 2017, Krueger et al., 2016). Almost all these studies consider decision-maker characteristics, vehicle attributes and government incentives when studying adoption decision, however, the impact of social influence is not brought to the foreground. The study by Bansal and Kockelman (2018) is one the few studies where social influence is explicitly considered in the timing of the adoption decision. Bansal and Kockelman (2018) asked respondent about potential purchase time in a multiple-choice question, where available choices are "never", "when 50% of friends adopted", "when 10% of friends adopted" and "as soon as available". They found that 39 percent of their sample would never purchase AVs, 32 per cent would purchase AVs when half of their friends have also purchased an AV, 15 per cent when 10% of their friends have purchased and 14 per cent as soon as AVs are available. This study reveals the significant impact of interpersonal social network in the adoption process. Nevertheless it does not capture the nuances and complexities of human interactions in depth. Eliciting the impact of social network influence on AV adoption is limited to aggregate values of market penetration, and although the substantial impact of communication channels is acknowledged (e.g. Talebian and Mishra, 2018) eliciting the effect on decision-makers' choice is not fully addressed.

This study fills the identified gap by designing a discrete choice experiment (DCE) to elicit the effect of communication channels on AV consideration. It is plausible to assume consumers will resort to external sources of information when deciding about AV adoption. We study various communication channels including mass media, social media, market penetration and WOM from social contacts. Participants are asked about their likelihood to consider AVs under various scenarios of received information (see Section 3). The impact of WOM is already studied (Talebian and Mishra, 2018), however this study distinguishes between WOM from hypothetical AV-owner and non-AV-owner. Furthermore, the survey includes a social network name generator component where respondents are asked to introduce their social contacts along with the contacts' characteristics, which is then used to customise the DCE based on the introduced contacts to make it more believable. This novel design, enables us to formulate how decision makers trust their social contacts, based on the characteristics of social contacts, and to elicit the taste towards trust when formulating AV consideration.

3. Survey instrument

The survey instrument in this study is a stated preference survey designed to elicit consumers' preferences towards AV with a focus on the received information about this innovation from various sources. Two overlapping streams of literature were reviewed for the survey design in this study. The first stream was about important attributes of AV from consumers' perspective, and the second stream was about the process of innovation diffusion and the influential sources of information when deciding to adopt to a new technology. The final survey instrument included five sections.

(1) **Social media and mass media engagement**. In this section respondents are asked about their social media habits, such as the platforms that they actively check and their frequency of use. Respondents are also asked about the frequency of checking news on TV channels, radio, newspapers, and online sources, and their popular news agencies.
(2) **Social network name generator**. Respondents are asked to introduce eight of their social contacts, the contacts' demographic attributes and their relationship with the contacts. The provided information in this section is used to customise the tasks in the DCE. The collected demographic attributes about contacts include gender, age, labour force status, and education. Introduced social contacts can be family members, social friends, colleagues, classmate, flatmate, neighbour, teacher, student, etc. When asked to introduce a sample of their social contacts, respondents are likely to list the contacts with whom they have a strong tie. This is while Granovetter (1977) showed weak ties play an essential role in the diffusion of innovation. To capture the impact of weak ties in our study and to increase the diversity of the introduced social contacts, respondents are first asked to introduce

four social contacts who "*are close to [them]*", then two contacts who "*used to be close to [them] in the past, but it has been nearly a year since [they] last contacted each other*", and then two contacts who "*are not very close to [them], but with whom [they] have frequent interactions due to other reasons*". Note that this classification is not meant to be used in the analysis but is put in place to increase the diversity and variety of introduced social contacts. This mechanism is needed to maintain a diverse list of social contacts including both strong and weak ties.

(3) **Preferences for AV**. The DCE of this study has eight hypothetical tasks where each task comprises two sections. The first section describes a hypothetical scenario of received informational cues about AVs from social contacts, mass media, social media platforms, and market penetration. Table 1 shows the list of attributes and their unique levels, and Fig. 1 shows a hypothetical scenario in this task. The received information from social contacts is presented as positive or negative feedback from the introduced social contacts in the previous section who may or may not have purchased an AV. For social media and mass media, the received sentiment is presented under four levels of *poor, fair, good, excellent* for social media, and *very negative, somewhat negative, somewhat positive, very positive* for mass media. Market penetration is defined as the proportion of new cars sold in the market that are AVs, and the attribute has four levels of 10, 20, 30 and 40 per cent. Similar to the DCE in the study by de Bekker-Grob et al. (2013) and Moylan et al. (2019), visual aids are provided to supplement the information provided under social media sentiments, mass media sentiments and market penetration so that it can be understood more easily. In this section, respondents are asked to indicate how likely they are to consider AV on a five-point Likert-scale.

In the second section of the DCE, four hypothetical vehicles are presented to the respondents based on their main attributes, and respondents are asked to specify the option they most prefer. We used a labelled DCE where two of the options are AV and two of the options are regular vehicles. Each option is further defined in terms of six attributes: body type, fuel type, driving range, refuelling/recharging time, operating cost, and purchase price. The details about this section of the DCE are excluded from the current paper as they are not directly related to the objectives of this study.

The values for attributes of both sections in each task were determined using an orthogonal design generated by NGENE software. This design consists of 120 rows in 15 blocks where each respondent faces 8 tasks.

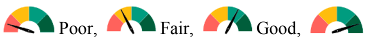
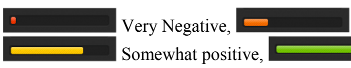
(4) **Interactions with social contacts.** This section includes questions about the quality and quantity of interactions with social contacts. Respondents are asked about number of interactions they had during the last month, and the main medium of communication with their social contacts introduced in section (2) of the survey. Respondents are also asked to rate their perception about technology savviness of their contacts, and the likelihood of following advices given or actions made by the contracts, using slider bar questions. The collected information in this section is used as the measurement component of the trust model (see Section 5).

(5) **Respondents' sociodemographic attributes.** Respondents are asked about their age, gender, income, education, employment status, vehicle ownership, residential tenure status and household structure.

4. Sample characteristics

The survey was conducted in December 2019, and our sample comprised residents in Sydney, Australia. The survey was administered online, and the assistance of a market research company was used for participant recruitment. The modelling sample of this

Table 1
The attributes and their corresponding levels in the choice task.

Attribute	Unique levels
WOM from AV-owners	• Of the eight friends and relatives that you have told us about, only PX has purchased an AV and recommends it • Of the eight friends and relatives that you have told us about, only PX has purchased an AV and **does not** recommend it • Of the eight friends and relatives that you have told us about, only PX and PY have purchased an AV. While PX recommends purchasing AVs PY **does not** recommend it • None of the friends and relatives that you have told us about have purchased an AV
WOM from Non-AV-owners	• Of those that have not purchased AV you have talked to PX. PX has not purchased an AV but recommends it • Of those that have not purchased AV you have talked to PX. PX has not purchased an AV and does not recommend it • There is no recommendation from those who have not purchased AV
Social Media	 Poor, Fair, Good, Excellent
Mass media	 Very Negative, Somewhat negative, Somewhat positive, Very positive
Market Penetration	 10 percent, 20 percent, 30 percent, 40 percent

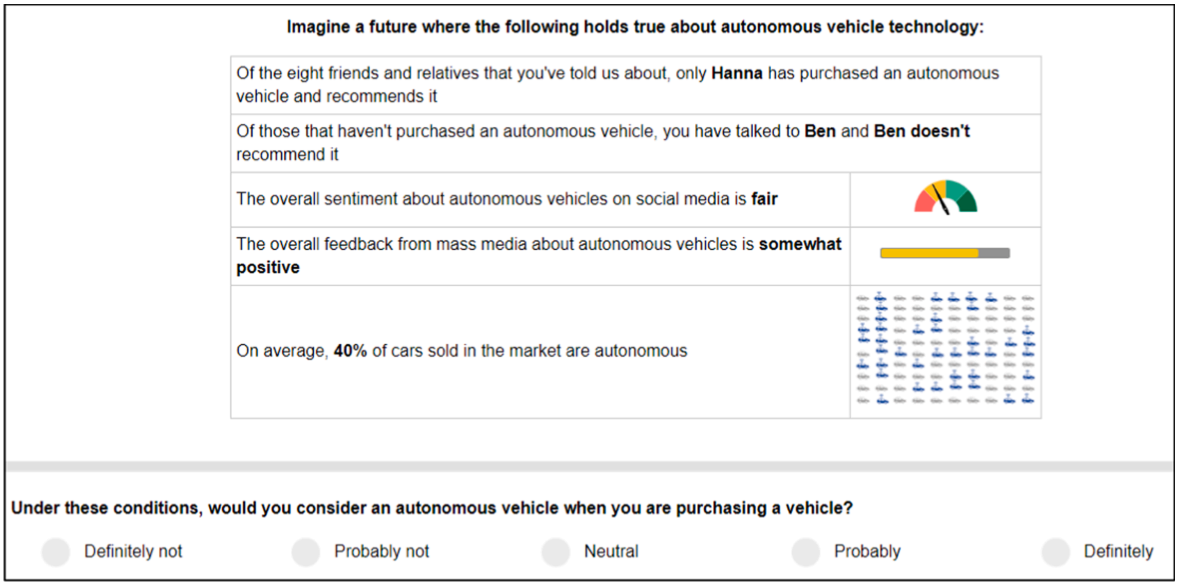


Fig. 1. Example Screenshot of a hypothetical scenario to elicit preference towards received information.

study contains information from 862 respondents. Table 2 presents the descriptive statistics for the key sociodemographic attributes of respondents, including age, gender, income, employment status, education, household structure, vehicle ownership, home type and tenure choice. Table 2 also presents the marginal distribution of each attribute for the collected sample and the population from the latest Census survey in 2016 (ABS, 2016). The gender distribution closely matches the population. For income and age, although the marginal distributions are close, the collected sample is slightly skewed towards older individuals with higher income. Our sample underrepresents full time employed individuals and overrepresents those who are not engaged in the labour force, and the education level for the respondents is slightly higher compared to the population. The portion of couples in the collected sample is lower than the population. In this study, we used an exogenous sampling method (Ben-Akiva and Lerman, 2018), therefore all the parameter estimates are unbiased despite the minor differences between the marginal distributions of sample and the population.

5. Econometric framework

A latent class choice model (LCCM) (see Vij and Walker, 2014) is used to segment the sample into relatively homogeneous classes when eliciting taste towards information channels. LCCM is a type of finite mixture discrete choice model that allows for discretisation of the population according to their characteristics and tastes which facilitates the interpretation of the observed behaviour within and across the segments. The LCCM in this study has three components: (i) a class membership model, (ii) a class-specific social contacts trust model, and (iii) a class-specific AV consideration model. Fig. 2 shows the structure of the proposed framework.

The class membership model in our study is formulated as a multinomial logit function shown in Eq. (1). In this equation, q_{ns} is a binary variable indicating whether individual n belongs to class s, z_n is a vector of characteristics for individual n and γ_s is vector of class-specific parameters indicating the sensitivity to the individual characteristics. S in this equation shows total number of classes which is determined based on a comparison across both statistical measures of fit and behavioural interpretation (see Section 6).

$$P(q_{ns}=1)=\frac{\exp(z_n\gamma_s)}{\sum_{s=1}^{S}\exp(z_n\gamma_s)} \tag{1}$$

The class-specific social contact trust model captures the level of trust that decision-makers put into their social contacts as feedback providers about their experience or knowledge about AVs. We consider trust to indicate the value of WOM from social contacts in decision-makers' eyes. Depending on how much decision-makers perceive their social contacts to be knowledgeable, trustworthy and technology savvy, the WOM from social contacts can have different values for the decision-makers. In this study, trust is a latent concept and cannot be directly measured. Therefore, to quantitatively capture this concept in the process of decision making, the trust model in this study is formulated with a structural component and a measurement component. The structural component is hypothesised to be a deterministic linear function of social contact characteristics, as shown in Eq. (2). The structural component of most integrated choice and latent variable (ICLV) models includes a random component as well (Ben-Akiva et al., 2002); however, the deterministic structural component adopted in this study will significantly reduce the parameter estimation burden. This simplifying assumption will result in a closed form likelihood function, as shown in the following paragraphs, therefore no simulation is required to estimate the parameters of the model. In this equation, τ_{nm} is the latent variable indicating the trust associated with social contact m by decision-maker n, z_{nm} is a vector of social contact characteristics, and α_s is a class-specific vector of parameters indicating sensitivity of trust towards characteristics.

Table 2
Descriptive statistics for the sample.

Variable	Sample		Population
	Number	Portion	Portion
Age			
between 18 and 29	174	20%	22%
between 30 and 49	304	35%	38%
between 50 and 69	262	30%	28%
70 years and over	122	14%	12%
Gender			
Female	423	49%	50.7%
Male	436	51%	49.3%
Income (average personal income per week)			
Negative income	3	0%	0.5%
Nil income	37	4%	8.5%
Between 1 and 149	37	4%	3.2%
Between 150 and 299	51	6%	6.4%
Between 300 and 399	72	8%	7.7%
Between 400 and 499	74	9%	7.4%
Between 500 and 649	68	8%	6.9%
Between 650 and 799	48	6%	7.2%
Between 800 and 999	84	10%	8.3%
Between 1000 and 1249	96	11%	8.9%
Between 1250 and 1499	65	8%	6.3%
Between 1500 and 1749	67	8%	5.5%
Between 1750 and 1999	51	6%	4.0%
Between 2000 and 2999	49	6%	6.4%
More than 3000	33	4%	4.7%
Not Stated	27	3%	8.2%
Employment			
Full time	351	41%	55.0%
Part time	167	19%	14.9%
Not Engaged	344	40%	30.1%
Education			
Postgraduate	124	14%	10.0%
Undergraduate	312	36%	31.4%
TAFE Certificate or equivalent	266	31%	46.0%
Other	160	19%	12.6%
Household Structure			
Couple with kids	347	40%	50.6%
Couple without kids	217	25%	32.7%
Single parent	53	6%	14.8%
Other	245	28%	1.9%
Household Vehicle Ownership			
No vehicle	65	8%	11.6%
1 vehicle	413	48%	38.5%
2 vehicles	305	35%	33.8%
3 + vehicles	79	9%	16.1%
Dwelling type			
House	605	70%	69.3%
Apartment	248	29%	29.8%
Other	9	1%	0.9%
Tenure type			
Owner	523	61%	63.8%
Renter	279	32%	34.4%
Other	60	7%	1.8%

$$\tau_{nm} = z_{nm}\alpha_s \tag{2}$$

In the measurement component of the trust model, the latent variable of trust is assumed to be measured by three indicators of Likert-scale survey questions about social contacts. As discussed in Section 3, after completing the tasks, respondents are asked to rate the social contacts they had listed earlier regarding three criteria: (i) how they perceive their social contacts to be technology savvy, (ii) how likely they are to follow advice from their social contacts and (iii) how likely they are to follow their social contact actions. Respondents rate their perception about the eight social contacts they introduced earlier in the survey under each of the three indicators, using slider bar questions. The continuous rates obtained from the slider questions were later aggregated into five discrete categories by dividing the range of the sliding bar into five equidistance sections.

The measurement component predicts the probability that respondent n chose option j (where $j = 1, \cdots, 5$) when rating social contact m. Assume v_{nmi} is the underlying latent construct behind the answer by decision-maker n to the indicator i when rating social contact m. We assume v_{nmi} can be defined by the trust that decision-maker n associates with the social contact m as shown in Eq. (3). In

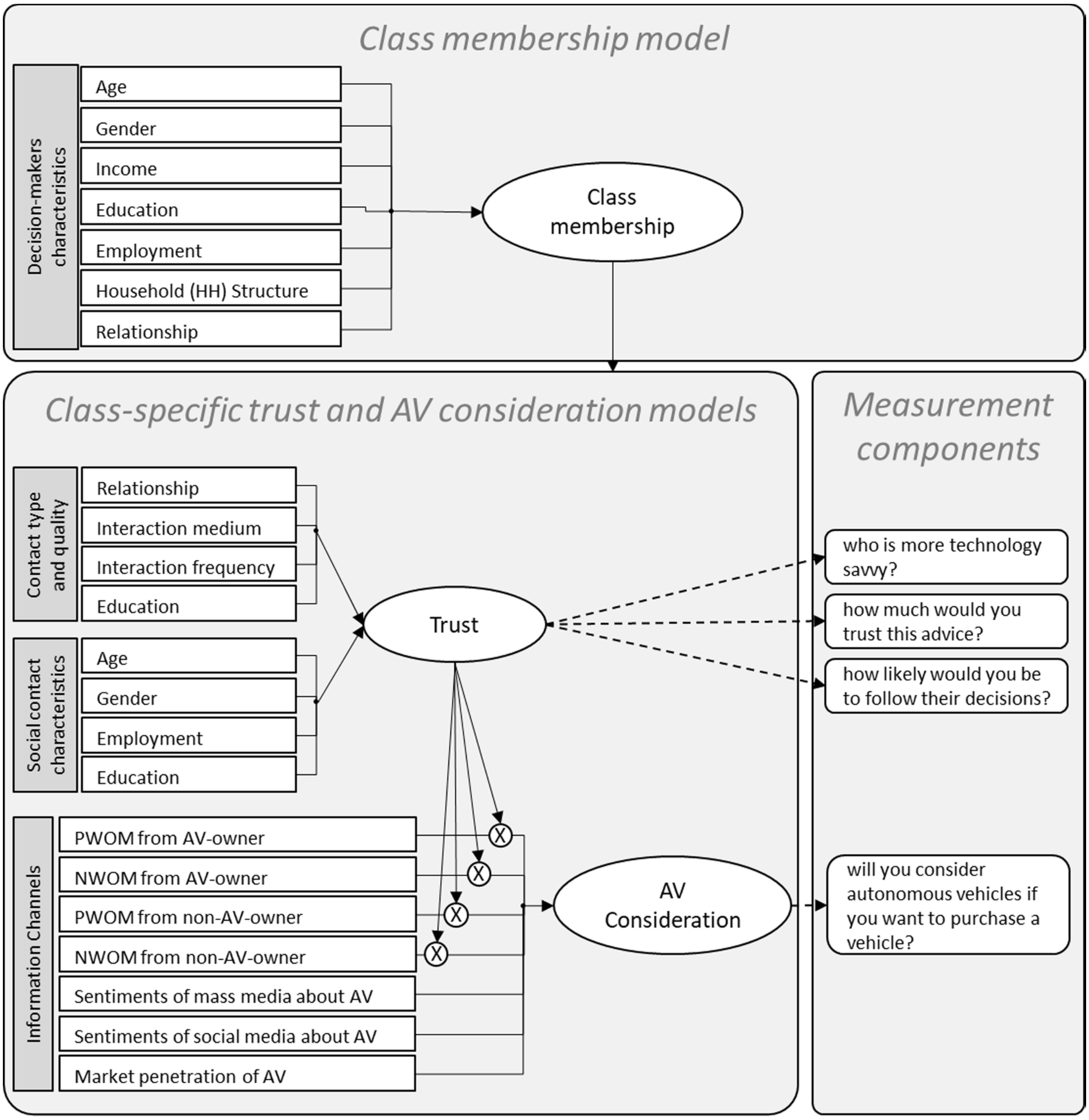


Fig. 2. Proposed econometric framework.

this equation ω_i is an indicator-specific parameter and η_{nmi} is the disturbance. For the sake of mathematical convenience, η_{nmi} is assumed to be independent and identically distributed (i.i.d) logistic with location zero and scale one across social contacts and decision-makers. Therefore, the measurement component takes the form of an ordered logit model and Eq. (4) shows the probability of decision-maker *n* selecting outcome *j* when rating social contact *m* on indicator *i*.

$$v_{nmi} = \omega_i \tau_{nm} + \eta_{nmi} \tag{3}$$

$$\begin{aligned} P\left(f_{nmji}=1|q_{ns}=1\right) &= P\left(\zeta_{j-1,i} \le v_{nmi} \le \zeta_{j,i}\right) = P\left(v_{nmi} \le \zeta_{j,i}\right) - P\left(v_{nmi} \le \zeta_{j-1,i}\right) = P\left(\eta_{nmi} \le \zeta_{j,i} - \omega_i\tau_{nm}\right) - P\left(\eta_{nmi} \le \zeta_{j-1,i} - \omega_i\tau_{nm}\right) \\ &= \frac{1}{1+\exp\left(\zeta_{j,i} - \omega_i\tau_{nm}\right)} - \frac{1}{1+\exp\left(\zeta_{j-1,i} - \omega_i\tau_{nm}\right)} \end{aligned} \tag{4}$$

In this equation f_{nmji} equals one if option *j* is selected for indicator *i*, and zero otherwise; and $\zeta_{j,i}$are indicator specific threshold parameters to be estimated, such that $\zeta_{j,i} \ge \zeta_{j-1,i}, \forall i = 1,2,3, \forall j = 1, \cdots, 5$ and $\zeta_{0,i} = -\infty \forall i = 1,2,3$ and $\zeta_{5,i} = +\infty \forall i = 1,2,3$.

Note that the measurement component of the framework is not class-specific, which is reflected in invariant threshold parameters across classes. If the thresholds are allowed to vary across the classes, then there would be no basis for comparing the variations in taste towards trust across the segments as the value of trust would be varying from class to class. As will be discussed in the following paragraph, AV consideration is assumed to be affected by WOM from social contacts interacted with the trust associated with them. The assumption of trust being invariant across latent segments enables us to compare the estimated values for the effect of trust (β^*) across the segments, which would not be possible if the thresholds were not hold invariant across segments.

The class-specific AV consideration model formulates the indicated level of AV consideration by respondents. As explained in Section 3, each respondent faces eight scenarios where each scenario presents a hypothetical situation of the received information about AV from various channels. Respondents are then asked to indicate how likely they are to consider an AV by selecting one of the five options from "definitely yes" to "definitely no". Let $u_{nt|s}$ denote the utility of considering AVs in scenario t for decision-maker n, conditional on the decision-maker belonging to class s. This study formulates $u_{nt|s}$ as a linear function of the received information plus a disturbance, as shown in Eq. (5).

$$u_{nt|s} = \boldsymbol{x}_{nt}\boldsymbol{\beta}_s + \boldsymbol{X}^*_{nt}\boldsymbol{\tau}_n\boldsymbol{\beta}^*_s + \epsilon_{nt|s} \tag{5}$$

In this equation, $\boldsymbol{x}_{nt}$ is a vector of explanatory variables indicating the informational cues from social media sentiments, mass media sentiments and market penetration of AVs to decision-maker n in task t, $\boldsymbol{\tau}_n$ is the vector of trust values associated with the eight social contacts for decision maker n, $\boldsymbol{X}^*_{nt}$ is a four by eight matrix indicating whether any of the four possible feedback of: "PWOM from AV-owner", "NWOM from AV-owner", "PWOM from non-AV-owner", and "NWOM from non-AV-owner" has been received from any of the eight introduced social-contacts. If the social contact m in task t falls into any of these categories, the corresponding binary variable in $\boldsymbol{X}^*_{nt}$ equals one, otherwise it equals zero. According to the design of the survey, for each social contact in each task at most one of the four binary variables can equal one. We argue that recommendations are not equally evaluated by the decision-makers and depending on the trust associated with the social-contacts providing the recommendation, its impact may vary. To account for this behaviour, $\boldsymbol{X}^*_{nt}$ is interacted by $\boldsymbol{\tau}_n$. $\boldsymbol{\beta}_s$ and $\boldsymbol{\beta}^*_s$ are class-specific vectors of parameters indicating sensitivity towards the informational cues, and $\epsilon_{nt|s}$ is the error term of the model. Similar to the measurement component of the trust model, $\epsilon_{nt|s}$ is assumed to be i.i.d logistic with location zero and scale one across decision-makers and tasks.

In each task, respondents are asked to reveal their level of consideration for the hypothesised scenario using a five level Likert-scale question. The AV consideration model predicts the probability that respondent n chose option j where $j = 1, \cdots, 5$. Given the assumed distribution for $\epsilon_{nt|s}$, this probability takes the form of an ordered logit model as shown in Eq. (6).

$$P\left(y_{ntj}=1|q_{ns}=1\right) = P\left(\xi_{j-1} \le u_{nt|s} \le \xi_j\right) = \frac{1}{1+\exp\left(\xi_j - \boldsymbol{x}_{nt}\boldsymbol{\beta}_s - \boldsymbol{X}^*_{nt}\boldsymbol{\tau}_n\boldsymbol{\beta}^*_s\right)} - \frac{1}{1+\exp\left(\xi_{j-1} - \boldsymbol{x}_{nt}\boldsymbol{\beta}_s - \boldsymbol{X}^*_{nt}\boldsymbol{\tau}_n\boldsymbol{\beta}^*_s\right)} \tag{6}$$

In this equation y_{ntj} equals one if decision-maker i selects option j in task t, and zero otherwise; and ξ_j are threshold parameters to be estimated, such that $\xi_j \ge \xi_{j-1}, \forall j = 1, \cdots, 5$ and $\xi_0 = -\infty$ and $\xi_5 = +\infty$.

In order to formulate the likelihood function, probabilities formulated in Eqs. (4) and (6) need to be wrapped up over social contacts and tasks respectively. In Eq. (7), probabilities in Eq. (4) are combined over social contacts, and measurement indicators to form a class-specific probability of observing the vectors of choices $\boldsymbol{f}_n = \langle f_{n,1,1,1}, \cdots, f_{n,8,5,3} \rangle$ as there are eight social contacts to rate against three indicators, and the rates are aggregated into five options.

$$P(\boldsymbol{f}_n|q_{ns}=1) = \prod_{m=1}^{8}\prod_{i=1}^{3}\prod_{j=1}^{5}\left[P\left(f_{nmji}=1|q_{ns}=1\right)\right]^{f_{nmji}} \tag{7}$$

Similarly, in Eq. (8), probabilities in Eq. (6) are combined over tasks to form a class-specific probability of observing the vectors of choices $\boldsymbol{y}_n = \langle y_{n,1,1}, \cdots, y_{n,8,5} \rangle$ as each decision-maker faces eight tasks with five available options.

$$P(\boldsymbol{y}_n|q_{ns}=1) = \prod_{t=1}^{8}\prod_{j=1}^{5}\left[P\left(y_{ntj}=1|q_{ns}=1\right)\right]^{y_{ntj}} \tag{8}$$

Eqs. (7) and (8) may be combined and marginalised over classes to yield the unconditional probability of the vectors of choices $\boldsymbol{f}_n$ and $\boldsymbol{y}_n$, then combined iteratively over decision-makers to form the likelihood function of this study shown in Eq. (9).

$$P(\boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\beta}^*, \boldsymbol{\gamma}|\boldsymbol{f}, \boldsymbol{y}, \boldsymbol{z}, \boldsymbol{x}, \boldsymbol{X}^*) = \prod_{n=1}^{N}\sum_{s=1}^{S}[P(\boldsymbol{y}_n|q_{ns}=1)P(\boldsymbol{f}_n|q_{ns}=1)]P(q_{ns}) \tag{9}$$

In this equation, N is total number of respondents which in this study is 862, and S refers to the number of segments which is to be estimated based on different measures of fit and interpretability of the results (see Section 6). All the unknown parameters of $\boldsymbol{\alpha}$, $\boldsymbol{\beta}$, $\boldsymbol{\beta}^*$ and $\boldsymbol{\gamma}$ are estimated using the software package PythonBiogeme (Bierlaire, 2016).

6. Results

6.1. Latent segmentation

The number of latent segments in the proposed model in Section 5 needs to be determined exogenously. To determine the number of segments, a single class model is developed, then in an iterative exploratory process number of classes are increased one by one, while three goodness-of-fit measures of Bayesian Information Criterion (BIC), Akaike Information Criterion (AIC) and Consistent AIC (CAIC), and the behavioural interpretability of the model are monitored. Table 3 summarises the results for thirteen iterations, where the search process was terminated when a decline in goodness-of-fit was observed across all three information criteria. During the process, on two occasions, the class-specific utility constant of AV consideration model for a specific segment tended towards negative infinity indicating that the decision-makers belonging to this segment show deterministic behaviour in that they are unwilling to consider AVs, regardless of the informational cues. For these segments, this behaviour is exogenously imposed by fixing the class-specific utility of AV consideration model to a large negative value. The minimum value of CAIC is observed for the eleven-class model and the minimum values of BIC and AIC occur at the twelve-class model, suggesting both specifications are acceptable. In this study the eleven-class model is selected as the preferred model specification because of better behavioural interpretability compared to the twelve-class model. Section 6.2 presents the parameter estimates for a single-class specification and Section 6.3 shows the results for the preferred eleven-class specification.

6.2. Single-class specification

This section presents the estimated parameters for a single class model. Table 4 shows the parameter estimates for the AV consideration and trust models in the single class specification. The single class specification does not require a class membership model. The parameter estimates for the AV consideration and trust models show the average teste in the sample towards the variables of the two models. According to Table 4, market penetration has a higher impact on AV consideration compared to mass media and social media sentiments. As expected, WOM has a higher impact when it is received from AV-owners. However, NWOM is not found significant when received from non-AV-owners.

According to the parameter estimates for the trust model, on average, females are trusted less, younger social-contacts are trusted more, those with full time or parttime job are trusted more than unemployed social-ties, and family members are trusted more. Regular interactions with social contacts on social media platforms, face-to-face interactions and communications over the phone are found to increase the trust, and frequency of interaction is a significant determinant in trust.

The parameter estimates for the single class specification reveal the average taste towards attributes. However, the average values are not an accurate reflection of individual decision-makers' behaviour when the sample is not homogenous. For instance, although the coefficient for NWOM from non-AV-owners is not found significant in Table 4, there may exist segments of the sample for whom this variable has a reducing impact on AV consideration. The eleven-class specification will address the taste heterogeneity in the sample and the rest of the discussions in this paper pertains to the parameter estimates for the eleven-segment specification.

6.3. Eleven-class specification

This section presents the estimated parameters for the final model specification. When presenting the results, the classes are ordered decreasingly according to the simulated AV consideration values (see Section 7.1).

Table 5 shows the parameter estimates for the class-specific AV consideration model ($\boldsymbol{\beta}$ and $\boldsymbol{\beta}^*$). All the attributes are found to be significant across at least one of the classes. It is expected that market penetration, social media sentiments, mass media sentiments and PWOM will have a positive impact on the utility, and NWOM will have a negative impact. If the estimated coefficient is not aligned with this expectation, then it is set to zero. It is noteworthy to mention that none of the constrained parameters are not statistically significant at 95 per cent confidence level in the unconstrainted model. The last two columns in this table pertain to the two segments for which a deterministic behaviour is identified. To exogenously impose the deterministic behaviour, the class-specific constants in these segments are set to −999 as a large negative value and the other parameters are set to zero. The simulated probabilities for AV consideration (see Section 7.1) confirms that −999 is a suitably large number to impose a deterministic behaviour.

As shown in Table 5, all informational cues are statistically significant in at least one of the classes. While NWOM was not significant in the single class model (Table 4), the latent segmentation practice shows that the sensitivity towards this variable is significant for members in classes 3 to 7. The parameter estimates in Table 5 are not comparable across classes due to dissimilar scale

Table 3
Goodness-of-fit measures versus number of classes.

Number of latent segmentations	Number of estimated parameters	Log-likelihood value	BIC	AIC	CAIC
1	38	−38572.2	77,255	77,220	77,293
2	73	−37154.7	74,522	74,455	74,595
3	106	−36449.0	73,207	73,110	73,313
4	143	−35752.9	71,922	71,792	72,065
5	180	−35412.5	71,349	71,185	71,529
6	217	−35044.7	70,721	70,523	70,938
7	254	−34712.5	70,165	69,933	70,419
8	291	−34492.9	69,833	69,568	70,124
9	328	−34346.8	69,649	69,350	69,977
10	365	−34218.0	69,499	69,166	69,864
11	394	−34130.3	69,408	69,049	**69,802**
12	431	−34061.8	**69,379**	**68,986**	69,810
13	468	−34029.8	69,422	68,996	69,890

Table 4
Parameter estimates, standard errors, and p-values for the AV consideration and trust models in the single-class specification.

Parameter	Coefficient	standard error	p-value
AV Consideration model			
Constant	0.960	0.104	0.000
Market Penetration	1.150	0.198	0.000
Mass Media	0.256	0.020	0.000
Social Media	0.205	0.020	0.000
WOM from AV-owner			
Positive	0.041	0.006	0.000
Negative	−0.019	0.009	0.030
WOM from non-AV-owner			
Positive	0.037	0.009	0.000
Negative	−0.009	0.009	0.280
Trust Model			
Constant	5.270	0.046	0.000
Gender is female	−0.282	0.024	0.000
Age			
Between 18 and 30	0.423	0.048	0.000
Between 30 and 50	0.399	0.045	0.000
Between 50 and 70	0.297	0.041	0.000
Employment			
Fulltime	0.358	0.034	0.000
Part time	0.264	0.041	0.000
Unemployed	−0.097	0.052	0.060
Relationship			
Child	0.482	0.051	0.000
Parent	0.075	0.054	0.170
Friend	0.082	0.030	0.010
Colleague	−0.076	0.053	0.150
Spouse or Partner	0.438	0.049	0.000
Interaction type			
Face to face	0.134	0.027	0.000
Phone	0.187	0.026	0.000
Social Media	0.228	0.028	0.000
Interaction Frequency is often and more	0.550	0.030	0.000

parameters. Accordingly, class-specific market elasticities are calculated in Section 6.4, and the findings are discussed in Section 7.

Table 6 presents the parameter estimates for the class-specific trust model ($\boldsymbol{\alpha}$). These parameters form the structural component of the model which formulates trust based on the demographics of social contacts, and the level of interaction between decision-makers and social contact as a measure of their tie strength. A positive coefficient in this table indicates an increasing impact from the variable on the class-specific trust, and a negative coefficient shows a decreasing effect from the variable on the class-specific trust. For example, females are less trusted in all classes, full time workers are trusted more in most of classes, and interaction on social media platforms will increase the trust. The class-specific trust behaviour is discussed in Section 7.1 and the general trends observed in this table are discussed in Section 7.2.

Table 7 presents the parameter estimates for the class membership model (γ). The demographic attributes in this table refers to the decision-maker characteristics. The effects coding is applied to preserve the identifiability of the model (Walker and Li, 2007, Bech and Gyrd-Hansen, 2005). We first constrained the coefficients in class 1 to zero, and after estimating the parameters of the model, we shifted the reference point so that the average value of the parameter estimates for each variable across all segments becomes zero. In this method, identifiability is preserved by constraining the summation of coefficients rather than constraining the coefficients in a particular class. The main benefit of this approach is the ease of model interpretation. In Table 7, a positive coefficient for a variable in a class indicates that members of that class are above the average with respect to that variable, and a negative coefficient shows that members of that class are below the average regarding that variable. For example, members in class 2 and class 5 are more likely to be younger than average, or members in classes 9 to 11 are likely to be less educated. The salient characteristics of the class members are discussed in Section 7.1 and the general trends of sociodemographic attributes across segments are discussed in Section 7.2.

6.4. *Class-specific elasticity of AV consideration*

The estimated parameters of the trust model (Table 6) and the estimated parameters of the class-membership model (Table 7) are comparable across segments. However, the parameters of the AV consideration model are not directly comparable due the varying scale parameter of the AV consideration model across segments. Therefore, in order to compare the effect of informational cues, we calculate class-specific market elasticities for each of the attributes.

Elasticities are calculated using arc elasticity. We used a sample enumeration to obtain the base market demand (denoted by D_0). Sum of the probabilities for “definitely” and “probably” in AV consideration are assumed to represent the demand for AV. It is noteworthy to mention that this is not a critical assumption as the focus is on the variation in demand, and not the demand itself. Then

Table 5
Parameter estimates (and standard error) for the class-specific AV consideration model.

Parameter	Class 1	Class 2	Class 3	Class 4	Class 5	Class 6	Class 7	Class 8	Class 9	Class 10	Class 11
Constant	5.97* (0.369)	5.45* (0.675)	1.15* (0.445)	2.48* (0.282)	1.09* (0.413)	1.13* (0.441)	−0.737* (0.279)	−0.584 (0.573)	−5.38* (1.27)	−999 (Fixed)	−999 (Fixed)
Market Penetration	1.11 (0.667)	2.61 (1.73)	1.86* (0.757)	1.93* (0.45)	0.000 (Fixed)	2.35* (0.739)	2.41* (0.527)	2.28 (1.2)	8.62* (3.02)	0.000 (Fixed)	0.000 (Fixed)
Mass Media	0.263* (0.067)	0.001 (0.181)	0.593* (0.083)	0.329* (0.05)	0.628* (0.097)	0.368* (0.074)	0.448* (0.056)	0.335* (0.123)	0.416 (0.255)	0.000 (Fixed)	0.000 (Fixed)
Social Media	0.181* (0.072)	0.000 (Fixed)	0.533* (0.077)	0.29* (0.044)	0.367* (0.093)	0.395* (0.075)	0.354* (0.052)	0.251* (0.123)	0.000 (Fixed)	0.000 (Fixed)	0.000 (Fixed)
WOM from AV-owner											
Positive	0.015 (0.02)	0.06 (0.047)	0.08* (0.019)	0.028* (0.014)	0.087* (0.026)	0.053* (0.027)	0.068* (0.016)	0.093 (0.056)	0.000 (Fixed)	0.000 (Fixed)	0.000 (Fixed)
Negative	−0.003 (0.029)	0.000 (Fixed)	−0.111* (0.027)	−0.037* (0.019)	−0.108* (0.037)	−0.136* (0.039)	−0.035 (0.023)	−0.041 (0.084)	−0.029 (0.116)	0.000 (Fixed)	0.000 (Fixed)
WOM from non-AV-owner											
Positive	0.047 (0.028)	0.000 (Fixed)	0.064* (0.025)	0.056* (0.018)	0.062 (0.034)	0.073* (0.033)	0.045* (0.022)	0.075 (0.086)	0.000 (Fixed)	0.000 (Fixed)	0.000 (Fixed)
Negative	−0.04 (0.026)	−0.019 (0.073)	−0.04 (0.027)	−0.025 (0.018)	0.000 (Fixed)	0.000 (Fixed)	−0.069* (0.021)	0.000 (Fixed)	−0.078 (0.129)	0.000 (Fixed)	0.000 (Fixed)

* Significant at 95% confidence level.

Table 6
Parameter estimates (and standard error) for the class-specific trust model.

Parameter	Class 1	Class 2	Class 3	Class 4	Class 5	Class 6	Class 7	Class 8	Class 9	Class 10	Class 11
Constant	6.42* (0.201)	−2.02 (1.51)	5.4* (0.224)	5.96* (0.091)	3.11* (0.206)	4.27* (0.184)	5.47* (0.103)	3.39* (0.269)	4.49* (0.145)	2.55* (0.325)	7* (0.301)
Female	−0.242* (0.079)	−0.054 (0.279)	−0.383* (0.122)	−0.288* (0.059)	−0.684* (0.123)	−0.445* (0.093)	−0.505* (0.066)	−1.07* (0.168)	−0.57* (0.121)	−0.951* (0.278)	−0.978* (0.211)
Age											
From 18 to30	0.08 (0.216)	4.72* (1.49)	1.42* (0.183)	0.326* (0.103)	0.789* (0.254)	0.666* (0.192)	0.134 (0.138)	0.098 (0.368)	0.052 (0.239)	0.484 (0.708)	1.45* (0.416)
From 30 to 50	0.1 (0.197)	4.84* (1.47)	1.66* (0.211)	0.11 (0.094)	1.4* (0.247)	0.519* (0.2)	0.154 (0.111)	0.33 (0.317)	0.499* (0.199)	0.534 (0.504)	0.856* (0.318)
From 50 to70	−0.16 (0.202)	5.46* (1.48)	1.04* (0.171)	0.11 (0.087)	1.19* (0.235)	0.456* (0.165)	0.193* (0.097)	0.738* (0.255)	0.145 (0.177)	0.122 (0.431)	−0.172 (0.269)
Bachelor or higher	−0.16 (0.202)	5.46* (1.48)	1.04* (0.171)	0.11 (0.087)	1.19* (0.235)	0.456* (0.165)	0.193* (0.097)	0.738* (0.255)	0.145 (0.177)	0.122 (0.431)	−0.172 (0.269)
Employment											
Fulltime	0.619* (0.135)	0.521 (0.389)	0.523* (0.126)	0.242* (0.074)	0.867* (0.165)	0.297* (0.133)	0.378* (0.082)	−0.128 (0.213)	0.634* (0.16)	0.443 (0.394)	−0.122 (0.264)
Part time	0.557* (0.148)	0.079 (0.447)	−0.217 (0.152)	0.123 (0.086)	0.179 (0.216)	0.338* (0.143)	0.317* (0.102)	−0.075 (0.268)	0.615* (0.22)	−0.144 (0.49)	0.906* (0.378)
Unemployed	0.471* (0.18)	−0.659 (0.517)	−0.869* (0.189)	−0.027 (0.098)	−0.36 (0.213)	−0.589* (0.219)	−0.126 (0.152)	−0.776 (0.406)	−0.041 (0.257)	−0.658 (0.584)	2.56* (0.629)
Relationship											
Child	0.034 (0.226)	0.289 (0.799)	0.642* (0.251)	0.356* (0.136)	1.06* (0.297)	0.522* (0.192)	0.465* (0.122)	0.903* (0.29)	0.959* (0.262)	0.802 (0.547)	−0.355 (0.46)
Parent	0.175 (0.174)	−1.1 (0.586)	0.425* (0.21)	−0.038 (0.102)	0.251 (0.233)	−0.004 (0.231)	−0.186 (0.152)	−0.564 (0.465)	0.042 (0.266)	0.063 (0.759)	0.586 (0.386)
Friend	−0.221* (0.108)	−0.548 (0.317)	0.139 (0.11)	0.054 (0.065)	0.276* (0.137)	0.083 (0.127)	0.311* (0.078)	−0.017 (0.197)	0.308* (0.144)	0.065 (0.321)	0.16 (0.256)
Colleague	−0.119 (0.175)	0.712 (0.468)	−0.681* (0.21)	0.072 (0.118)	−0.241 (0.366)	0.369* (0.169)	−0.112 (0.173)	−0.361 (0.378)	−0.019 (0.222)	0.201 (0.854)	0.313 (0.449)
Spouse or Partner	0.448* (0.136)	−0.456 (0.553)	1.52* (0.269)	0.069 (0.1)	1.34* (0.266)	0.823* (0.189)	0.23 (0.132)	0.258 (0.331)	0.782* (0.248)	0.543 (0.566)	0.174 (0.558)
Interaction medium											
Face to face	−0.049 (0.091)	2.69* (0.346)	0.871* (0.11)	0.07 (0.058)	0.205 (0.137)	−0.022 (0.101)	0.303* (0.075)	−0.07 (0.185)	0.627* (0.13)	0.916* (0.367)	−0.47* (0.22)
Phone	−0.178* (0.093)	3.85* (0.347)	0.313* (0.111)	−0.024 (0.064)	−0.163 (0.151)	0.3* (0.093)	0.55* (0.069)	0.732* (0.181)	0.117 (0.134)	0.368 (0.295)	0.48* (0.232)
Social Media	0.039 (0.09)	3.01* (0.334)	0.551* (0.127)	0.196* (0.063)	0.86* (0.14)	0.398* (0.098)	0.404* (0.076)	0.579* (0.199)	0.732* (0.164)	1.08* (0.373)	−0.83* (0.268)
Interaction frequency is often and more	0.25* (0.093)	−0.77* (0.398)	0.573* (0.12)	0.469* (0.063)	3.02* (0.169)	0.239* (0.11)	0.652* (0.081)	0.507* (0.199)	0.021 (0.156)	−0.147 (0.32)	0.096 (0.226)

* Significant at 95% confidence level.

Table 7
Parameter estimates (and standard error) for the class membership model.

Coefficient	Class 1	Class 2	Class 3	Class 4	Class 5	Class 6	Class 7	Class 8	Class 9	Class 10	Class 11
Constant	1.05 (Fixed)	0.573* (1.54)	0.208* (0.902)	2.36 (0.661)	1.43 (0.875)	−0.63* (1.08)	−0.43* (0.753)	−1.81* (1.21)	−0.8* (0.987)	−0.31* (1.26)	−1.64* (1.46)
Age	−2.745 (Fixed)	−5.325* (3.19)	1.225 (1.75)	−1.065 (1.25)	−3.925* (1.75)	0.155 (1.47)	2.975 (1.33)	2.385 (2.01)	3.405 (1.74)	1.645 (2.38)	1.265 (2.49)
Female	−0.316 (Fixed)	−0.578* (0.69)	0.804 (0.385)	−0.001 (0.311)	1.044 (0.432)	−0.039 (0.366)	0.814 (0.325)	−0.397* (0.509)	−0.329* (0.443)	−1.376* (0.717)	0.375 (0.614)
Bachelor or higher	1.139 (Fixed)	−0.131* (0.679)	−0.131* (0.399)	0.281* (0.319)	−0.361* (0.423)	0.257* (0.368)	0.119* (0.332)	0.623* (0.5)	−0.551* (0.457)	−1.201* (0.723)	−0.041* (0.622)
Employment											
Fulltime	0.544 (Fixed)	1.468 (1.43)	−0.195* (0.633)	−0.09* (0.526)	0.451* (0.664)	1.32 (0.864)	−0.158* (0.549)	−0.926* (0.914)	−0.065* (0.735)	−1.326* (1.1)	−1.026* (1.1)
Part time	0.172 (Fixed)	1.362 (1.38)	−0.701* (0.628)	0.109* (0.515)	0.404 (0.623)	1.272 (0.869)	−0.559* (0.553)	−0.21* (0.896)	−1.378* (0.799)	−0.248* (1.02)	−0.224* (1.08)
Retired	0.102 (Fixed)	1.282 (2)	0.031* (0.785)	−0.293* (0.689)	0.299 (0.889)	0.991 (0.974)	−0.635* (0.682)	0.307 (0.993)	−1.888* (0.852)	−0.176* (1.1)	−0.019* (1.2)
Family Structure											
Couple No kids	0.331 (Fixed)	−0.939* (1.28)	0.808 (0.569)	0.012* (0.498)	0.148* (0.618)	0.6 (0.561)	0.812 (0.513)	0.802 (0.773)	1.441 (0.641)	−2.179* (1.43)	−1.839* (1.15)
Couple with kids	0.294 (Fixed)	−0.478* (1.07)	0.752 (0.587)	−0.218* (0.475)	−0.647* (0.64)	0.616 (0.551)	0.51 (0.516)	1.177 (0.767)	0.094* (0.713)	−0.469* (1.1)	−1.626* (1.05)
Income higher than 800	−0.25 (Fixed)	−0.238* (0.86)	−0.715* (0.472)	−0.041 (0.378)	−0.827* (0.499)	0.248 (0.449)	−0.196 (0.393)	0.498 (0.557)	−0.104 (0.568)	0.394 (0.722)	1.23 (0.732)
In Relationship	0.418 (Fixed)	0.277* (1.07)	−0.842* (0.513)	0.091* (0.448)	1.024 (0.584)	−0.544* (0.5)	−0.185* (0.459)	−1.092* (0.692)	−0.463* (0.605)	0.373* (1.09)	0.939 (0.977)

* Significant at 95% confidence level.

the attribute of interest is increased by 1 percent and AV demand is recalculated (denoted by D_1). Eq. (10) shows the elasticity for the continuous attributes. In this equation, D shows the demand and x denotes the attribute.

For the binary variables of WOM, the percentage change in arc elasticity is calculated relative to the mid-point. We first set the binary variable for all respondents equal to zero ($x_0 = 0$) and calculated market demand (denoted by D_0') then we set the binary variable equal to one ($x_1 = 1$) for the entire sample and recalculated the demand (D_1'). Eq. (11) shows the market elasticity for the binary attributes.

$$E_x = \frac{\%changeinD}{\%changeinx} = \frac{\partial D/D}{\partial x/x} = 100 \times \frac{(D_1 - D_0)}{D_0} \tag{10}$$

$$E_x = \frac{\%changeinD}{\%changeinx} = \frac{D_1' - D_0'/(D_1' + D_0')/2}{x_1 - x_0/(x_1 + x_0)/2} = \frac{D_1' - D_0'}{D_1' + D_0'} \tag{11}$$

To complete the sample enumeration trust values are needed. Trust can be estimated using the structural model (Eq. (2)), or it can be directly obtained from the indicators (inferred from Eqs. (3) and (4)). In this study we inferred trust from the indicators and parameter estimates for the thresholds and indicators' weights. In this process, for every introduced social contact, a likelihood function is set up as Eq. (12). In this equation $\mathscr{L}_{nm}$ is the likelihood function of respondent n and social contact m, which is a function of trust associated with m by n (denoted by τ_{nm}). The likelihood function is defined as the probability of selecting the chosen level (j^*) on the three indicators (i). The maximum likelihood estimator is used to estimate the value of trust. The process is repeated for all respondents and all social contacts.

$$\mathscr{L}_{nm}(\tau_{nm}) = \prod_{i=1}^{3} P\left(f_{nmj^*i} = 1\right) = \prod_{i=1}^{3} P\left(\zeta_{j^*-1,i} \le \omega_i \tau_{nm} \le \zeta_{j^*,i}\right) \tag{12}$$

The calculated elasticity values are summarised in Table 8. The values in this table are comparable across the segments. A general trend observed in this table is the increase in elasticities as we move from left to right. This observation indicates that the effect of informational cues is less for those individuals who are highly likely to consider AVs, and the effect increases for those who are less likely to consider AVs. The former group are more determined in considering AVs and informational cues marginally influence their decision, whereas the latter have not formed a strong persuasion towards AVs and their decision is more influenced by the received information.

7. Discussion

7.1. Class-specific behavioural interpretations

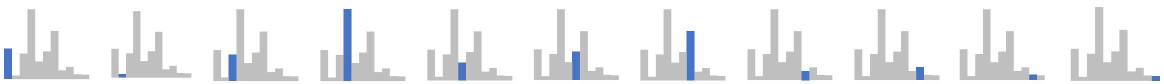
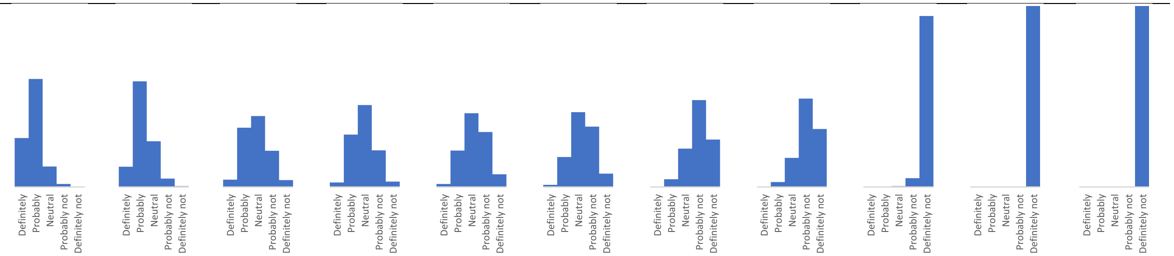
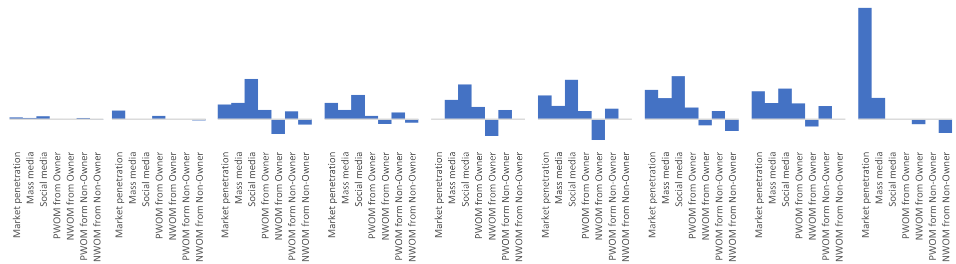
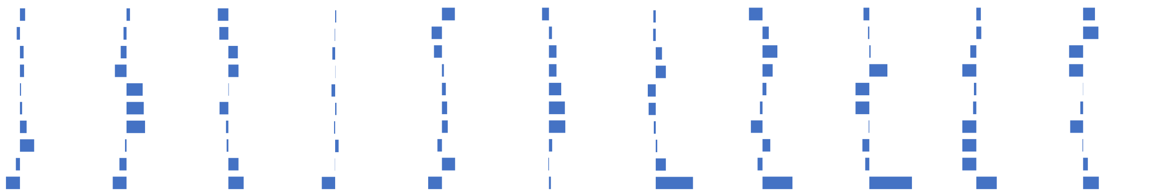
This section highlights the salient attributes of each segment and provides information on how identified segments differ from one another. The differences are discussed according to decision-makers' demographics found significant in the class membership model, the significant parameters in the class-specific trust model, and the effect of informational cues in the class-specific AV consideration model. Table 9 presents the salient characteristics for the eleven identified segments in the final specification. The probabilities of AV consideration are calculated from sample enumeration, using the estimated parameters and the collected information from respondents. The segments are then sorted according to the simulated AV consideration for presentation purposes. In the sample enumeration, the shares of each segment are also calculated using the estimated membership function. The class membership shares, and the distributions of AV consideration are presented in the top two rows of Table 9. The class-specific elasticity values in this table are obtained from Table 8. The salient attributes in the trust model are determined according to the statistically significant parameter estimates in Table 6, and the bar charts in the demography row are the visualisation of the parameter estimates in the class-membership model in Table 7.

Class 1 comprises 11.91 per cent of the sample and is most likely to consider AVs. 27.1 per cent of the individuals belonging to this class will "definitely" consider AVs and 59.8 per cent will "probably" consider AVs. The elasticity towards informational cues have their minimum in this class. This observation shows individuals in this class have already formed a tendency towards AVs, and the received information does not influence their decision much. They trust their employed contacts more and higher level of interaction

Table 8
Class-specific market elasticity of the parameters in the AV consideration model.

Parameter	Class 1	Class 2	Class 3	Class 4	Class 5	Class 6	Class 7	Class 8	Class 9	Class 10	Class 11
Market Penetration	0.03	0.17	0.29	0.32	–	0.47	0.57	0.55	2.18	–	–
Mass Media	0.02	–	0.32	0.18	0.38	0.26	0.41	0.31	0.42	–	–
Social Media	0.05	0.00	0.78	0.47	0.68	0.77	0.84	0.60	–	–	–
WOM from AV-owner											
Positive	0.01	0.07	0.18	0.07	0.24	0.16	0.23	0.31	–	–	–
Negative	0.00	–	−0.30	−0.10	−0.33	−0.41	−0.12	−0.14	−0.10	–	–
WOM from non-AV-owner											
Positive	0.02	–	0.15	0.13	0.18	0.21	0.15	0.25	–	–	–
Negative	−0.02	−0.03	−0.11	−0.07	–	–	−0.23	–	−0.27	–	–

Table 9
Prominent features of the segments.

Attribute	Class 1	Class 2	Class 3	Class 4	Class 5	Class 6	Class 7	Class 8	Class 9	Class 10	Class 11
Class membership shares	11.91%	1.47%	10.09%	27.51%	6.93%	10.98%	18.98%	3.47%	4.89%	1.95%	1.82%
											
Distribution of simulated AV consideration											
Elasticity (Table 8)										-	-
Attribute	Class 1	Class 2	Class 3	Class 4	Class 5	Class 6	Class 7	Class 8	Class 9	Class 10	Class 11
Salient attributes in the trust model (Table 6)	Employed; contact frequency	Age; medium of contact	Age; partner; face to face interaction	Male	Age, family members, frequency of contact	Unemployme; frequency of contact	Interaction over the phone	Age, male child,	Age	Face to face interaction, social media interaction	Young age, not social media interaction
Demography (Class membership model) (Table 7) *Married* *Income* *Couple with kids* *Couple without kids* *Retired* *Part time* *Fulltime* *Bachelors* *Female* *Age*											

increases their trust. Individuals most likely to belong to this class are employed young individuals with college education.

Class 2 has the smallest market share of 1.47 per cent. The probabilities of “definitely” and “probably” considering AV in this class are calculated as 11.2 and 58.4 per cent, respectively. The AV consideration for individuals in this class is influenced by market penetration and PWOM from AV-owners. The trust for individuals in this class is highly dependent on age, education, and the medium of contact. Individuals in class 2 have the highest rate of fulltime and part time employment, and they form the youngest cohort.

Class 3 comprises 10.09 per cent of the sample. The most frequent level of AV consideration in this class is “neutral” with 39.2 per cent. Although compared to class 1 and 2, members in this class show lower AV consideration, the distribution is still skewed towards considering AVs, with 36.9 per cent “definitely” or “probably” considering AVs. In this class, social media sentiments have the largest positive effect on AV consideration, followed by mass media sentiments and market penetration, and NWOM from AV-owners has the largest negative effect. Individuals in class 3 associate higher trust with their partner or spouse and those with whom they have face to face interaction. Individuals in class 3 are likely to be females who live as couples and are older compared to average.

Class 4 with 27.51 per cent has the highest share in the sample. Nearly 45 per cent of individuals in this class are “neutral” in AV consideration and 31.5 per cent “definitely” or “probably” consider AV. Similar to class 3, social media sentiments have the largest positive effect and NWOM from AV-owners has the largest negative effect on AV consideration. In this class, frequency of interaction has the largest positive effect on trust, and being female has the largest negative effect. The most likely individuals in class 4 are younger individuals.

In all the segments from class 5 to class 11, which forms 49.02 per cent of the sample, the simulated distribution is skewed towards not considering AV. Accordingly, to demonstrate the behavioural differences in these segments, we shift the focus to the probabilities of not considering AVs in the following paragraphs. In class 5, it is 37.5 per cent likely for individuals to “probably” or “definitely” not consider AV. Similar to class 3 and 4, social media sentiments have the largest positive effect on AV consideration and NWOM from AV-owners has the largest negative effect. The likely trusted social contacts in this class are family members, middle aged and those with whom individuals have frequent contacts. Most likely individuals belonging to class 5 are young married females with income below the average.

Class 6 comprises 10.98 per cent of the sample. The probability of “definitely” or “probably” not considering AV for this class is 40.8 per cent. In class 6, NWOM from AV owners has the largest negative effect on AV consideration among all segments. Regarding the trust component, unemployment has the highest negative effect on trust. Demographics of individuals belonging to class 6 are more likely to be employed and not in relationship.

Class 7 is the second largest segment with 18.98 per cent of the sample. 48.1 per cent of individuals belonging to this class “probably”, and 26.3 percent “definitely” do not consider AV. Among all segments, social media sentiments have the highest positive effect on AV consideration in class 7. The other salient feature is the higher negative effect of NWOM from non-AV-owners compared to AV-owners. The distinct feature of trust in class 7 is that interaction over phone has higher effect compared to other media. The likely individuals in this class are older females who are not retired.

Class 8 comprises 3.47 per cent of the sample. In this class, the likelihoods of “probably” and “definitely” not considering AV are 48.2 and 32.1 per cent. The effect of market penetration on AV consideration in this class is almost as high as social media sentiments. In this class, older social contacts, and males are more likely to be trusted. The positive and significant coefficient for *child* in Table 6 indicates that members in class 8 will associate a higher trust for their children, accordingly, the effect of WOM will be higher when received from their children. Individuals belonging to this class are older than average and more likely to have a household structure of couple with kids.

In class 9 with 4.89 per cent of the sample, the dominant choice for AV consideration is “definitely not” with the probability of 94.5 per cent. In this class, the highest positive effect on AV consideration is from market penetration followed by mass media sentiments. Unlike previous segments, AV consideration is not influenced by social media sentiments. The trust model shows employed middle-aged male social contacts are more likely to be trusted in this class. The most likely demographics of individuals belonging to this class are old age and no college education.

The deterministic behaviour of “definitely” not considering AV is the prominent feature of class 10 and class 11. The two classes combined comprise 3.78 per cent of the sample. The most likely demographics of individuals belonging to these classes are no college education, not employed, and older age. However, individuals in class 11 are more likely to be married and have higher incomes compared to individuals in class 10. Regarding the trust component, interacting on social media increases trust for individuals in class 10, but it has a decreasing effect in class 11.

7.2. General trends

Our findings suggest a positive effect from social media and mass media sentiments on AV consideration. This is aligned with the findings from previous studies which acknowledge a significant impact from social media and mass media on the market uptake of products (e.g. Tuarob and Tucker, 2014, Srinivasan et al., 2009, Axsen et al., 2013). To the best of our knowledge this paper is the first study to explore this impact for AV consideration. The results show social media sentiment has the largest positive effect in seven segments out of the eleven identified segments which nearly comprises 90 percent of the sample. This underpins the significance of this study in exploring the impact of emerging communication channels in the era of digital communication.

The results confirm findings from previous studies that higher market penetration increases the likelihood of considering AV (Lavasani, 2016, Talebian and Mishra, 2018). This is because high market penetration of an innovation (such as AVs) can be considered as a non-verbal confirmation signal from the community (Rogers, 2010). Furthermore, high market penetration can constitute a new social norm which can consequently induce a desire for conformity in the decision-makers.

One of the main contributions of this study is eliciting taste towards PWOM and NWOM for AV consideration. Our findings show that the previous finding of PWOM having a larger effect compared to NWOM (East et al., 2016) does not hold true for all the segments. Also, depending on whether WOM is received from an AV-owner or a non-AV-owner the effect may be different. For instance, across classes 3 to 6, NWOM from AV-owner has a higher effect on AV consideration compared to PWOM from AV-owners, whereas this ranking is different for WOM from non-AV-owner.

Going from left to right in Table 9, along with a reduction in the likelihood of AV consideration, several general trends are observed. Regarding decision-makers' attributes, higher age, no college education, and unemployment are the most likely demographics in segments with lower AV consideration likelihood. On the other hand, individuals belonging to class 1 and 2 are more likely to be male. Our findings are consistent with previous studies that younger individuals are less concerned in adopting AVs (Schoettle and Sivak, 2014b), individuals with higher level of education have higher tendency towards AVs (Haboucha et al., 2017) and men are more open to the technology than women (Schoettle and Sivak, 2014b). However, the latent segmentation in this study reveals a dominantly male niche of the population who will not consider AVs regardless of informational cues.

In the trust model, a noticeable pattern across all the segments is that females are trusted less compared to males. Also, social contacts who are educated or are employed in a fulltime position are trusted more. Family members are shown to be trusted more compared to other categories of social contacts, and interactions on social media increases trust in almost all segments.

8. Conclusion

This study examined the impact of communication channels on AV consideration with a focus on digital communication platforms and recommendations from social network. The level of trust associated with social contacts is added to the model to control the effect of word-of-mouth. The data was obtained from a stated preference survey from a representative sample of 862 participants from Sydney, Australia. To minimise hypothetical bias, the discrete choice experiment was customised according to the introduced social contacts by the participants. The collected data was analysed using a latent class choice model. The proposed econometric framework consists of a class membership model, a class-specific AV consideration model, a class-specific trust model, and two measurement components. The results revealed eleven segments in the sample.

According to the results, social media sentiment has the largest positive effect on AV consideration for 90 percent of the sample. In most classes, NWOM from AV-owners has a higher effect compared to NWOM from non-AV-owners. We showed that NWOM does not always have a higher impact compared to PWOM and the ranking varies from segment to segment. The estimated parameters for the trust model showed that family members are trusted more compared to other social contacts. Also, men, and contacts with fulltime job and college education are trusted more. Furthermore, interactions with social contacts on social media have a positive impact on trust. According to the results, higher age, no college education, and unemployment are the most likely demographics in segments with lower AV consideration. Men are more likely to belong to segments with higher AV consideration.

This paper investigates the impact from social influence on the evolution of AV preferences. We studied the effect of word-of-mouth from social contacts and we adjusted the effect according to the trust associated with social contacts, but when measuring trust, we overlooked social behaviours such as modesty, vanity, snobbism, virtue signalling and status seeking. Therefore, one obvious research direction is to include these behavioural nuances when evaluating interpersonal interactions in a social system. Moreover, we explored the influence exerted by mass media and social media sentiments on AV consideration; however, we did not consider the effect of fractured information and echo chamber formation in these platforms. While these phenomena may not directly affect the sensitivity towards mass media and social media, there may be correlations between positive or negative views towards AVs and media consumption habits. Finally, the dataset of this study is obtained from a stated preference survey. While stated preference and speculative data are useful in early stages of the innovation diffusion process, the conclusions drawn from these sources are limited as many respondents are unfamiliar with AVs, therefore they differ in their ability to hypothesise their perception from the survey questions. Hence, a logical next step to continue this study is combining it with pilot and trial studies on AVs to achieve more credible responses.

Authors statement

M.G. and A.V. conceived of the presented idea. M.G. and A.V. developed the theory and designed the survey. M.G. collected the data. M.G. and A.V. performed the computations and developed the models. M.G. and A.V. discussed the results and contributed to the final paper.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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